expand by at least a factor of 10^{30}, an amount that is readily achieved in specific realizations of inflationary expansion, all the regions in space that we currently see—all the regions in space whose temperatures we have measured—were able to communicate as easily as the adjacent kitchen and living room and hence efficiently come to a common temperature in the earliest moments of the universe.[16] In a nutshell, space expands slowly enough in the very beginning for a uniform temperature to be broadly established and then, through an intense burst of ever more rapid expansion, the universe makes up for the sluggish start and widely disperses nearby regions.

That's how inflationary cosmology explains the otherwise mysterious uniformity of the microwave background radiation suffusing space.

Inflation and the Flatness Problem

A second problem addressed by inflationary cosmology has to do with the shape of space. In Chapter 8, we imposed the criterion of uniform spatial symmetry and found three ways in which the fabric of space can curve. Resorting to our two-dimensional visualizations, the possibilities are positive curvature (shaped like the surface of a ball), negative curvature (saddle-shaped), and zero curvature (shaped like an infinite flat tabletop or like a finite-sized video game screen). Since the early days of general relativity, physicists have realized that the total matter and energy in each volume of space—the *matter/energy density*—determine the curvature of space. If the matter/energy density is high, space will pull back on itself in the shape of a sphere; that is, there will be positive curvature. If the matter/energy density is low, space will flare outward like a saddle; that is, there will be negative curvature. Or, as mentioned in the last chapter, for a very special amount of matter/energy density—the critical density, equal to the mass of about five hydrogen atoms (about 10^{-23} grams) in each cubic meter—space will lie just between these two extremes, and will be perfectly flat: that is, there will be no curvature.

Now for the puzzle.